26900122 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCND301 - PAEDIATRIC NUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Identify the component of the TICLS framework which assesses a child's muscle tone and spontaneous movement? [K] / [CO1]

   a). Tone b). Interactiveness

   c). Look/Gaze d). Speech/Cry

2. Identify the major factors associated with nutritional deficiencies in children with CP. [K] / [CO2]

   a). Increased physical activity

   b). Increased food variety

   c). Increased nutritional requirement

   d). Poor oral-motor coordination, swallowing dysfunction, and GER

3. Identify the hormone-related pathway affected by undernutrition and involved in normal growth. [K] / [CO3]

a). Growth hormone–IGF-1 axis b). Insulin-glucagon axis

c). Antidiuretic hormone d). Thyroid-stimulating hormone axis

4. Identify the major urinary abnormality characteristic of nephrotic syndrome [K] / [CO4]

a). Hematuria b). Heavy proteinuria

c). Glycosuria d). Ketonuria

5. State the main purpose of the Specific Carbohydrate Diet. [U] / [CO2]

a). To increase sucrose intake

b). To increase grain consumption

c). To restore and maintain a healthy gut microbial population

d). To increase lactose intake

6. State the purpose of early identification of ADHD. [U] / [CO3]

a). To improve outcomes through early recognition and management

b). To completely prevent ADHD

c). To eliminate the need for treatment

d). To avoid educational support

7. Identify the body fluids most commonly used for biochemical assessment of nutritional status. [K] / [CO1]

a). Saliva and sweat b). Blood and urine

c). Cerebrospinal fluid and saliva d). Lymph and sweat

8. Identify the common metabolic complication of pediatric obesity? [U] / [CO3]

a). Hypoglycemia b). Vitamin A deficiency c). Type 2 diabetes mellitus d). Iron deficiency anemia

9. A child is prescribed a classical ketogenic diet with a 4:1 ratio. Which meal composition best represents this ratio? [A] / [CO4]

a). 4 g carbohydrate : 1 g fat

b). 4 g fat : 1 g protein + carbohydrate

c). 4 g protein : 1 g fat

d). 4 g fat : 1 g protein

10. Name the metabolic complication involving abnormal acid-base balance in CKD. [K] / [CO4]

a). Metabolic acidosis b). Respiratory alkalosis

c). Metabolic alkalosis d). Respiratory acidosis

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. Describe the clinical manifestation of iodine deficiency [U] / [CO1]

12. State the role of a small intestinal biopsy in diagnosing celiac disease. [K] / [CO2]

13. List any four common food allergens. [K] / [CO3]

14. Explain how defective CFTR function results in thick, sticky secretions. [U] / [CO4]

15. Define epilepsy. [K] / [CO4]

16. Describe the procedure for recording and plotting measurements on a WHO Growth Chart. [U] / [CO1]

17. Explain gastroesophageal reflux in preterm infants and mention any two management measures. [U] / [CO2]

[2x5 = 10]

Part C

Short Essay

Answer any **6** questions

Each question carries **5** marks

18. Define nutritional assessment in peadiatrics and explain any two parameters of anthropometry. [K] / [CO1]

19. Explain the types of cyanotic congenital heart disease with suitable examples. [U] / [CO2]

20. Describe the different types of dental caries seen in children. [K] / [CO3]

21. Describe the diagnosis of diarrhoea in children. [K] / [CO4]

22. Describe the different assistive feeding devices used for children with developmental disabilities and explain how they promote safe and independent feeding. [U] / [CO2]

23. Examine the short term consequences of undernutrition. [An] / [CO3]

24. Describe the laboratory investigations used in the diagnosis of T1DM. [K] / [CO4]

25. Define lactose intolerance and explain any two methods used to diagnose lactose intolerance. [U] / [CO2]

[5x6 = 30]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26. Evaluate the role of specialised diets used in the management of food allergy, with reference to elimination diets and other dietary management approaches, nutritional monitoring and regular follow-up. [E] / [CO3]

27. Explain the mineral and vitamin requirements in preterm and low birth weight infants. [U] / [CO2]

28. Explain the different standards used for interpreting anthropometric data in children, with reference to IAP, Gomez, and NCHS standards. [U] / [CO1]

29. Explain the nutritional management of celiac disease in children, including common nutritional deficiencies, nutrient requirements, and appropriate dietary modifications. [U] / [CO4]

[10x2 = 20]